



STM32F3 Technical Training

For reference only

Refer to the latest documents for details



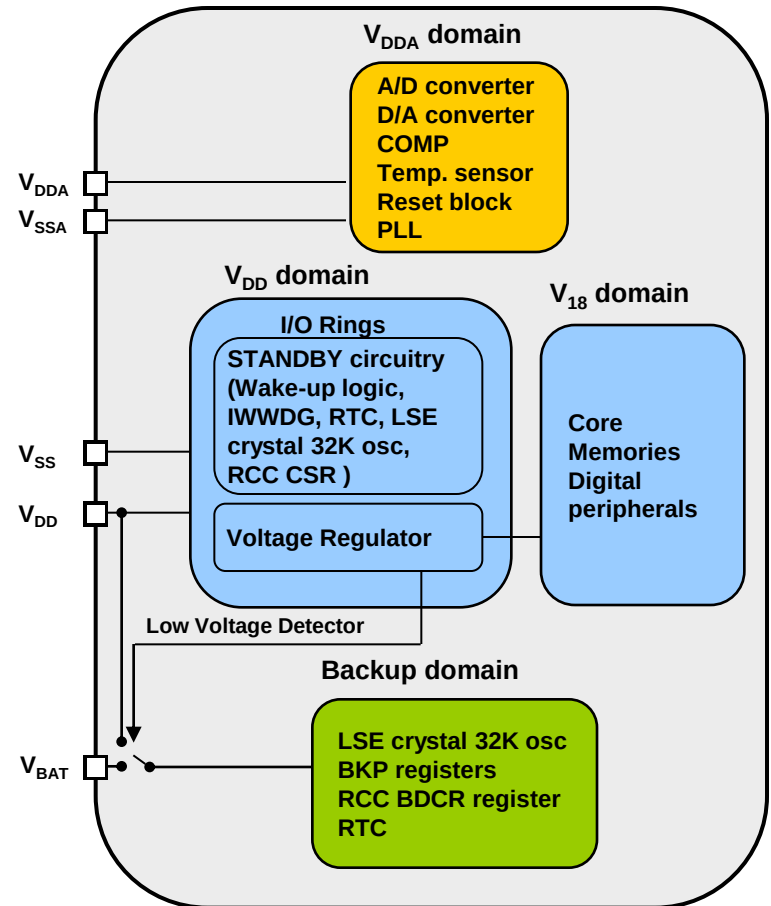
Power Control (PWR)

SRM32F30x Power Supply

3

Power Supply Schemes

- **VDD = 2.0 to 3.6 V : External Power Supply for I/Os (or VDD = 1.8 +/- 8%: %: External Power Supply for I/Os with internal regulator is OFF.)**
- **VDDA = 2.0 to 3.6 V: External Analog Power supplies for ADC,DAC, Reset blocks, RCs and PLL.**
 - ➔ DAC working only if VDDA >=2.4 V
- **VBAT = 1.65V to 3.6 V: For Backup domain when VDD is not present.**
- **Power pins connection:**
 - VDD and VDDA can be provided by a separated power supply source.
 - VSS and VSSA must be tight to ground

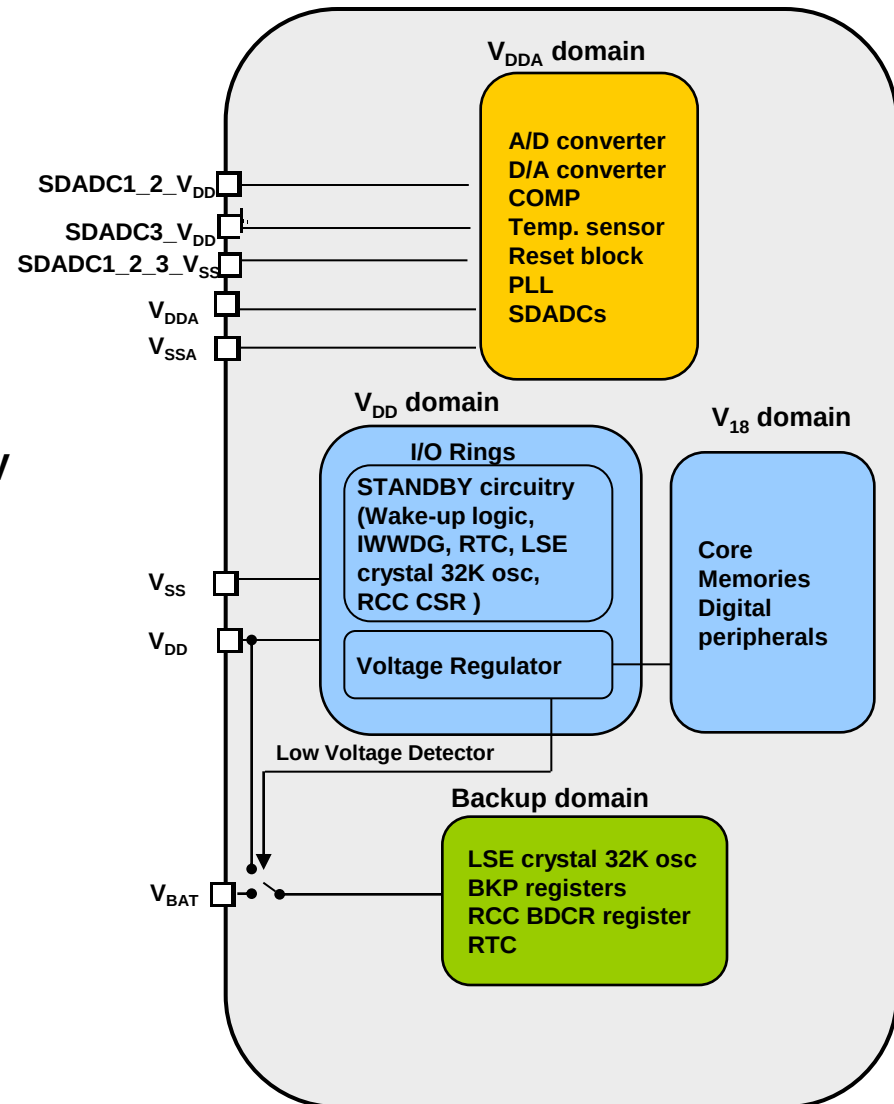


SRM32F37x Power Supply

4

Power Supply Schemes

- **VDD = 2.0 to 3.6 V : External Power Supply for I/Os (or VDD = 1.8 +/- 8%: %: External Power Supply for I/Os with internal regulator is OFF.)**
- **VDDA = 2.0 to 3.6 V: External Analog Power supplies for ADC,DAC, Reset blocks, RCs and PLL.**
 - ➔ **ADC and DAC working only if VDDA >=2.4 V**
- **VBAT = 1.65V to 3.6 V: For Backup domain when VDD is not present.**
- **SDADCx_VDD = 2.2 to 3.6V : External Analog Power supplies for SDADCs with:**
- **Power pins connection:**
 - VDD and VDDA can be provided by a separated power supply source.
 - VSS, VSSA and SDADCx_VSS must be tight to ground
 - The SD1_SD2_VDD and SD3_VDD can be different from VDD , VDDA and from one another.



Power Sequence

5

- When VDD power supply source is different from VDDA power supply source ($VDD < VDDA$)
 - The VDDA voltage level must be always greater or equal to the VDD voltage
 - During power-on, the VDDA must be provided first (before VDD)
 - During power-off, it is allowed to have temporarily $VDD > VDDA$, but the voltage difference must be $< 0.4V$
 - could be maintained by an external Schottky diode
- When SDADCx power supply is different from VDDA, VDD power supply and from one another:
 - $SDADCx_VDD \leq VDDA$
 - $SDADC1_VDD / SDADC2_VDD \leq SDADC3_VDD$
 - $SDADC3_VDD$ must start before or at the same time as $SD12_VDD$

Supply monitoring and Reset circuitry

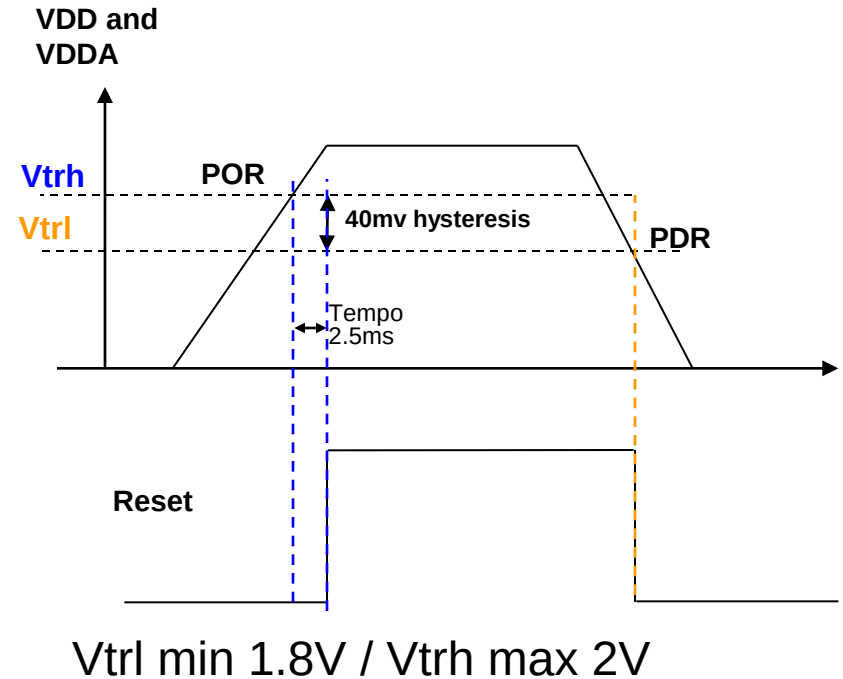
6

- The STM32F3xx POR / PDR circuitries are always active and monitor two supply voltages: VDD and VDDA.
- The POR supervisor circuit monitors only **VDD**
- The PDR supervisor circuit monitors **VDD** and **VDDA**
 - The PDR supervisor on VDDA can be disabled by programming Option byte.

Power On Reset / Power Down Reset

7

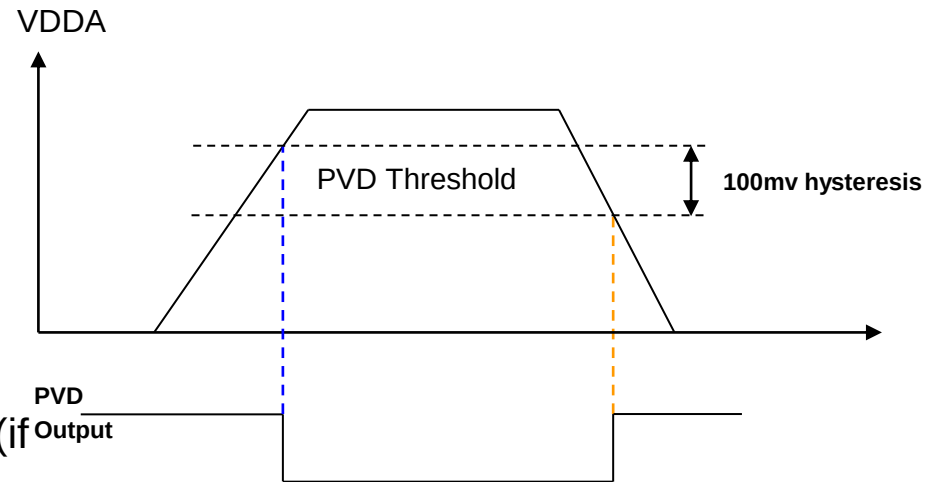
- Two Integrated POR / PDR circuitries guarantees proper product reset when voltage is not in the product guaranteed voltage range (2V to 3.6V)
 - No need for external reset circuit**
- POR and PDR have a typical hysteresis of 40mV
- The PDR detector monitors VDD and also VDDA (if kept enabled in the option bytes). The POR detector monitors only VDD.



Programmable Voltage Detector (PVD)

8

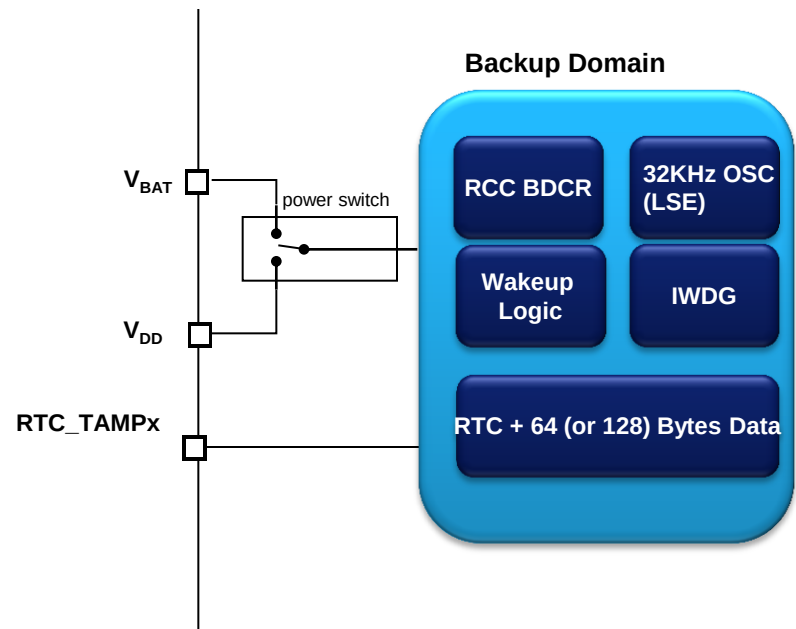
- Programmable Voltage Detector
 - Enabled by software
 - Monitors the VDDA power supply by comparing it to a threshold
 - Threshold configurable from 2.1V to 2.9V by step of 90mV
 - Generates interrupt through EXTI Line16 (if enabled) when $VDDA < \text{Threshold}$ and/or $VDDA > \text{Threshold}$
 - ➔ Can be used to generate a warning message and/or put the MCU into a safe state



Backup Domain

9

- Backup Domain contains
 - Low power calendar RTC (Alarm, periodic wakeup from Stop/Standby)
 - 64 and 128 Bytes Data RTC registers in STM32F30x and STM32F37x respectively.
 - Separate 32kHz Osc (LSE) for RTC
 - RCC BDCR register: RTC clock source selection and enable + LSE config
 - → Reset only by RTC domain RESET
- VBAT independent voltage supply
 - Automatic switch-over to VBAT when VDD goes lower than PDR level
 - No current sunk on VBAT when VDD present.
- 2 x Tamper events detection: resets all user backup registers
- TimeStamp event detection.



Low Power Modes (1/4)

10

- **SLEEP Mode:** Core stopped, peripherals kept running
 - Entered by executing special instructions
 - **WFI** (Wait For Interrupt)
 - Exit: any peripheral interrupt acknowledged by the Nested Vectored Interrupt Controller (NVIC)
 - **WFE** (Wait For Event)
 - An event can be an interrupt enabled in the peripheral control register but NOT in the NVIC or an EXTI line configured in event mode
 - Exit: as soon as the event occurs → No time wasted in interrupt entry/exit
 - Two mechanisms to enter this mode
 - **Sleep Now:** MCU enters SLEEP mode as soon as WFI/WFE instruction are executed
 - **Sleep on Exit:** MCU enters SLEEP mode as soon as it exits the lowest priority ISR
 - To further reduce power consumption you can save power of unused peripherals by gating their clock

Low Power Modes (2/4)

11

- **STOP Mode:** all peripherals' clocks, PLL, HSI and HSE are disabled, SRAM and registers contents are preserved.
 - If the RTC and IWDG are running, they are not stopped in STOP (either as their clock sources)
 - To further reduce power consumption, the Voltage Regulator can be put in Low Power mode
 - Wake-up sources:
 - WFI was used for entry: any EXTI Line configured in Interrupt mode (the corresponding EXTI Interrupt vector must be enabled in the NVIC)
 - WFE was used for entry: any EXTI Line configured in event mode
 - EXTI line source can be: one of the 16 external lines, PVD output, RTC alarm, **COMPx**, **I2Cx**, **USARTx** or the **CEC** (*).
 - The **I2Cx**, **USARTx**, **CEC** (*) can be configured to enable the HSI RC oscillator for processing incoming data. If this is used, the voltage regulator should not be put in the low-power mode but kept in normal mode.
- ➔ After resuming from STOP the clock configuration returns to its reset state (HSI used as system clock).

(*): CEC is available in STM32F37x only.

Low Power Modes (3/4)

12

- **STANDBY Mode:** Voltage Regulator off, the entire V18 domain is powered off.
 - SRAM and register contents are lost except registers in the Backup domain and STANDBY circuitry
 - PLL, the HSI RC and the HSE crystal oscillators are also switched off.
 - RTC and IWDG are kept running in STANDBY (if enabled)
 - In STANDBY mode all IO pins are high impedance and non-active except:
 - Reset pad (still available)
 - RTC pins (if configured)
 - PC14 & PC15 could be forced to output high/low in RTC registers
 - WKUPx pins (if enabled)
 - **Wake-up sources:**
 - WKUPx pins rising edge
 - RTC alarm and tamper events
 - External reset in NRST pin
 - IWDG reset
- ➔ After wake-up from STANDBY mode, program execution will restart in the same way as after a RESET.

STM32F3xx Low Power modes

13

Mode name	Entry	Wakeup	Effect on 1.8V domain clocks	Effect on VDD domain clocks	Voltage regulator	IO state	Wakeup latency
SLEEP, SLEEP now or SLEEP on-exit	WFI	Any interrupt	CPU CLK OFF no effect on other clocks or analog clock sources	None	ON	All I/O pins keep the same state as in the Run mode	None
	WFE	Wake-up event					
STOP	PDDS, LPSSDR bits + SLEEPDEEP bit + WFI or WFE	Any EXTI line (configured in the EXTI registers, <u>internal</u> and <u>external</u> lines)	All 1.8V domain clocks OFF	HSI and HSE and oscillators OFF	ON, in low power mode (depending on PWR_CR)	all I/O pins are high impedance (*)	HSI RC wakeup time + regulator wakeup time from Low-power mode
STANDBY	PDDS bit + SLEEPDEEP bit + WFI or WFE	WKUP pin rising edge, RTC alarm, RTC tamper event, external reset in NRST pin, IWDG reset			OFF		Reset phase

(*): **Standby mode:** all I/O pins are high impedance except:

- Reset pad (still available)
- RTC pins PC14 and PC15 if configured in the RTC registers.
- WKUP pin 1 (PA0) and WKUP pin 2(PC13), if enabled.

- How many power supply domains are available?

- What is the power sequence recommendation?

- What are the wake-up sources from STOP mode?



Thank you !

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